# Statistical watermarking for LLM text

## Detection under paraphrasing and translation

Research problem, working prototype, evidence and next experiments

Project progress review 21 October 2026

# The problem statement

## Can a signal inserted during generation remain detectable after the wording changes?

### Insertion

A cooperating generator adds a keyed pattern to token choices.

### Transformation

A user rewords, translates, or mixes the passage with other text.

### Detection

The detector measures evidence for that particular pattern.

Scope: attribution to a deliberately embedded signal

# Why this baseline?

## A simple mechanism we can reproduce and stress-test

| Property | Why it helps this project |
| --- | --- |
| No model retraining | Adds a bias while the model selects its next token |
| Independent detection | Uses the tokenizer and key without generation weights |
| Adjustable strength | Exposes a measurable tradeoff with writing quality |
| Known research baseline | Supports comparison with published alternatives |

Baseline selection is a methodological choice, not a robustness guarantee

# How our watermark works

| During generation | During detection |
| --- | --- |
| Key + previous token determine the favored set | The same key and context rebuild the favored set |
| About 25% of eligible choices are favored | Count favored choices in eligible token pairs |
| A bias makes favored choices more likely | Compare the observed count with chance |

**Same key. Same tokenizer. Same partition rule.**

# What the signal score means

## z = (G - gamma x T) / sqrt(T x gamma x (1 - gamma))

| G | Favored choices counted |
| --- | --- |
| T | Eligible token pairs checked |
| gamma | Favored share, approximately 0.25 |

## Our experimental cutoff is z >= 4

A detector score expresses evidence for a pattern, not authorship certainty

# Research foundations

| Paper | Contribution | Implication for us |
| --- | --- | --- |
| Kirchenbauer et al. ICML 2023 [R1] | Green-list promotion and statistical detection | Reproducible insertion and detection baseline |
| Kirchenbauer et al. ICLR 2024 [R2] | Rewriting, residual fragments and longer text | Robustness depends on attack and observed length |
| Sadasivan et al. TMLR [R3] | Stress-tests detectors with recursive paraphrasing | Evaluate deliberate evasion and false positives |

The papers study different settings. Our example cannot settle all of them.

# Research directions beyond the baseline

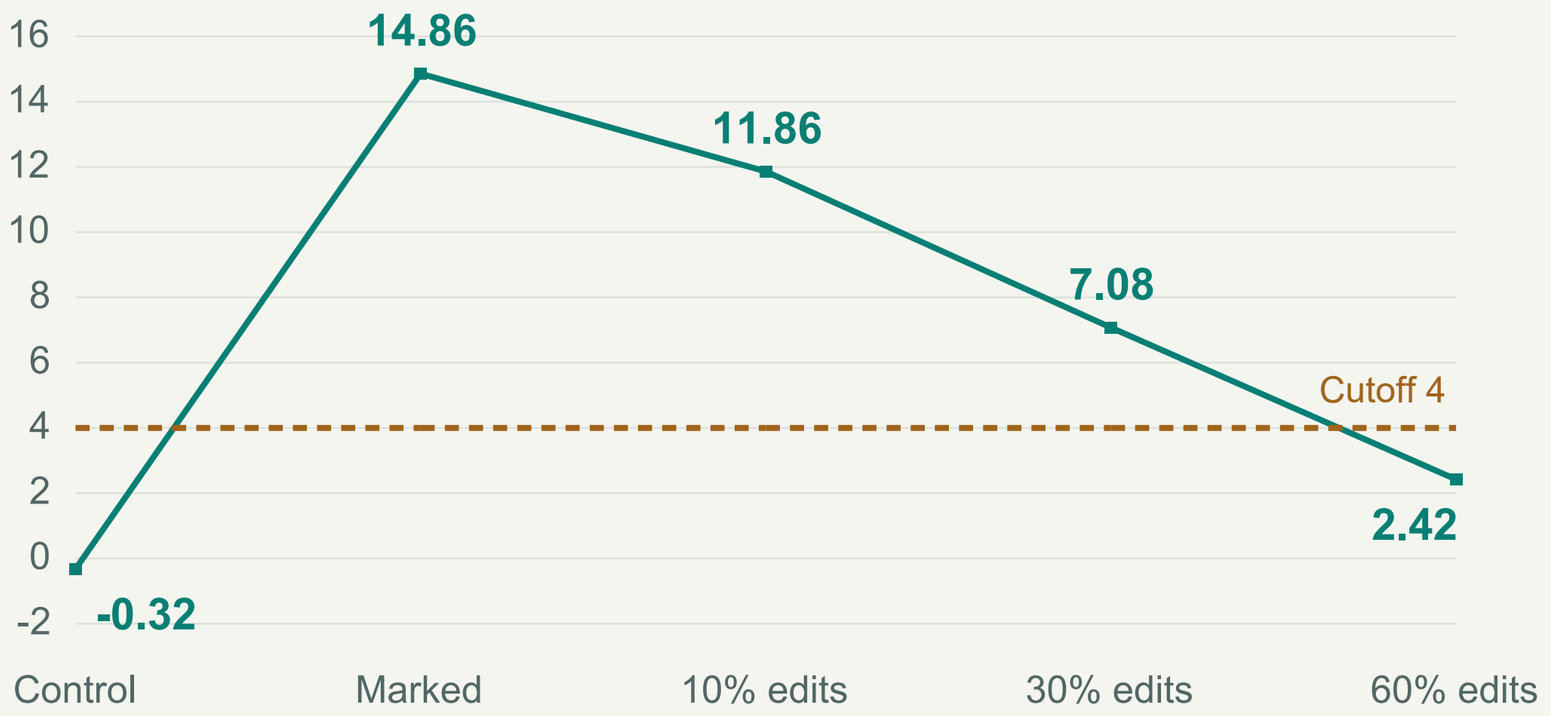
| Approach | Research contribution | Our next comparison |
| --- | --- | --- |
| Distortion-free [R4] | Keyed sampling with alignment-based detection | Detection power and edit tolerance |
| Semantic invariant [R5] | Semantic context informs the watermark | Paraphrase tolerance and added cost |
| SynthID-Text [R6] | Scalable watermarking in production settings | Quality, detection and latency tradeoffs |
| WaterBench [R7] | Joint evaluation across tasks and lengths | A fair and reproducible benchmark |

# What we have built

| Component | Current evidence |
| --- | --- |
| Keyed insertion and detector | Implemented and exercised with real generation |
| Readable paired passages | Qwen2.5-0.5B-Instruct runs locally |
| Three recursive paraphrases | Saved natural-language example with per-round scores |
| Paragraph comparison interface | Complete prose, optional highlights and exports |
| Batch study and back-translation | Code exists. Comprehensive measured study remains pending |

Working software and research validation are tracked separately

# Synthetic integration evidence



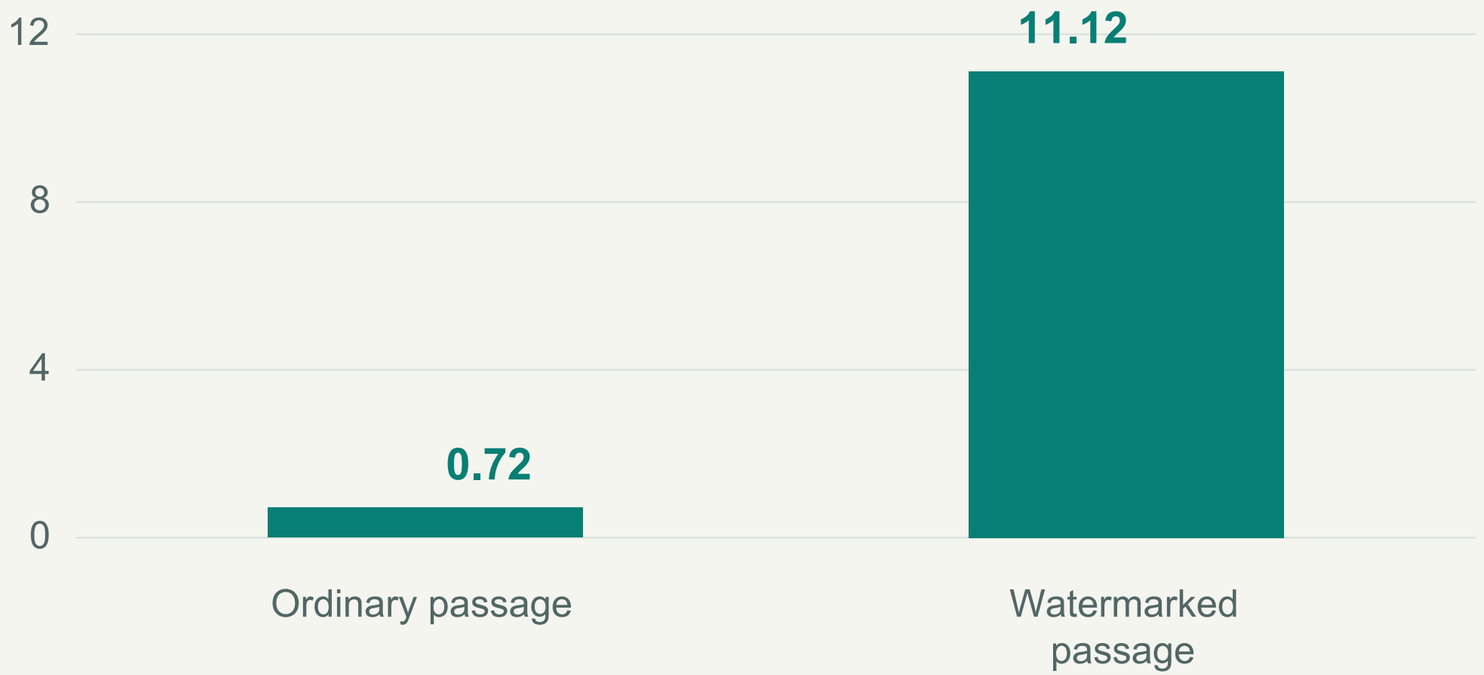

## 20 seeds

**Detections at z >= 4**

Control 0 / 20
Marked 20 / 20
10% edits 20 / 20
30% edits 20 / 20
60% edits 2 / 20

SYNTHETIC TEST ONLY: random token replacement is not semantic rewriting

# Real text: insertion and detection work




**0.72**

Ordinary passage

**11.12**

Watermarked passage

One paired example. Experimental detection cutoff: 4.

# The writing changes during paraphrasing

## Original watermarked passage

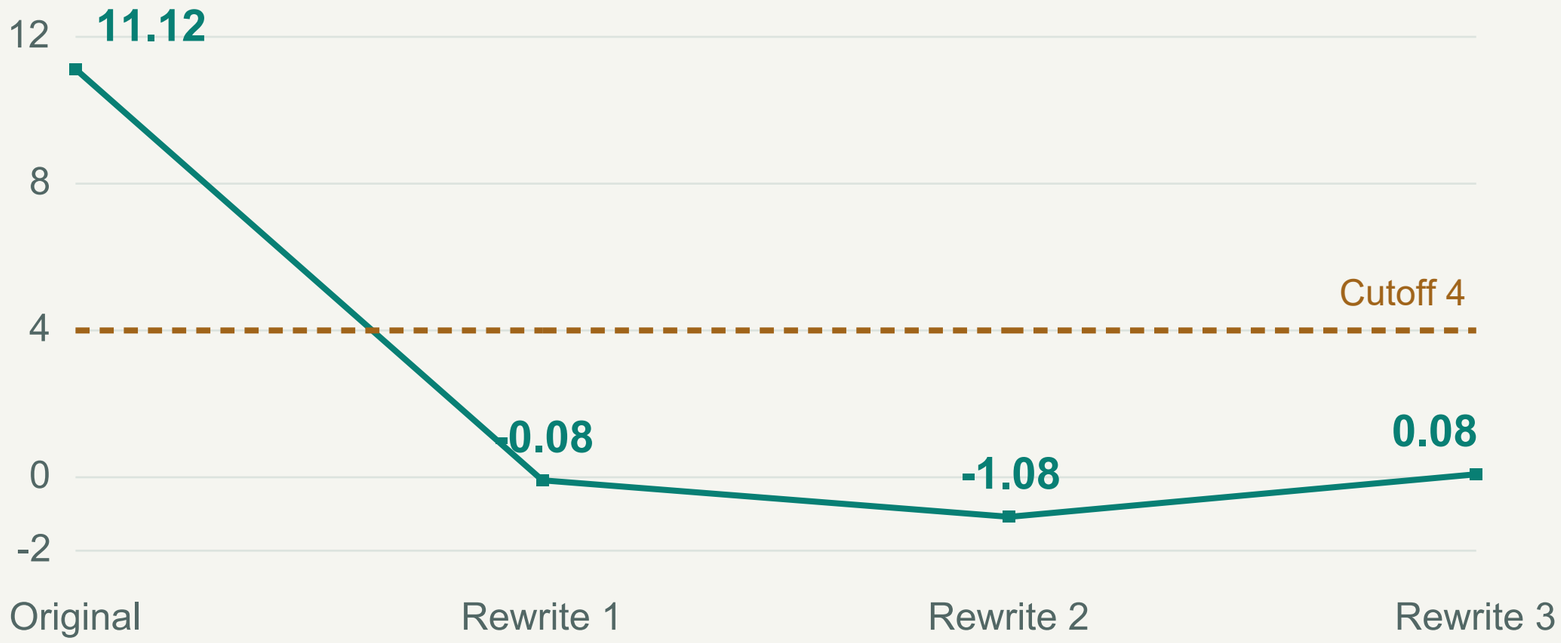
Urban gardens play an essential role in improving the quality of life for residents and contributing to the sustainability of cities. These spaces provide a natural habitat that not only beautifies our neighborhoods but also serve as microclimates, offering shading, fresh air, and greenery where people can relax, work and socialize.

## First paraphrase

Urban gardens significantly contribute to the overall well-being of communities through numerous positive impacts. Their primary function is to improve the aesthetic value of our neighborhoods while concurrently serving crucial ecological roles such as pollination, water conservation, and carbon sequestration.

Exact excerpts. Similar topic does not establish preserved meaning or factual accuracy.

# One rewrite removed detectable evidence




**59.5% to 24.7%**

Favored choices
Original vs first rewrite

**195 vs 194**

Checked token pairs
Nearly equal evidence length

Local example only. Cutoff = 4. Values rounded for display. Negative scores are preserved.

# What this result actually establishes

## A demonstrated weakness in the current baseline

### Supported

The original signal is detectable.
This paraphrase chain falls below the cutoff.

### Still unproven

Robustness across topics and models.
A calibrated low false-positive rate.

A missed watermark is not evidence of human authorship

# Limits of the current evidence

| Limitation | Why it changes the conclusion |
| --- | --- |
| One local paraphrase chain | An example cannot estimate a general survival rate |
| Small model and limited text lengths | Results may change with model, task and output length |
| Meaning and completion quality unchecked | Semantic drift and truncation can distort attack evaluation |
| False-positive calibration incomplete | A chosen cutoff does not establish an empirical error rate |
| Translation benchmark incomplete | Implemented translation code is not measured robustness |

# Next controlled experiment

## 20 prompts x 3 seeds x 6 conditions

360 scored texts in the proposed pilot

| Condition | Purpose |
| --- | --- |
| Ordinary control and original watermark | Measure insertion, detection and accidental triggers |
| Recursive paraphrases 1, 2 and 3 | Measure signal decay and meaning preservation |
| English-French-English translation | Measure a different transformation pathway |

# Further research and comparisons

| Research question | Experiment to run |
| --- | --- |
| Can stronger bias help without harming prose? | Sweep delta and gamma with blinded quality review |
| Does more text recover residual evidence? | Study longer passages and windowed detection |
| Can a different signal survive rewording? | Compare semantic-context and alignment methods |
| Does robustness generalize? | Use independent paraphrasers, domains and languages |
| What does deployment cost? | Measure generation latency, detection time and memory |

# Expected outcomes and deliverables

| Deliverable | Evidence required before completion |
| --- | --- |
| Reproducible experiment package | Versioned configuration, complete passages and CSV results |
| Robustness report | Detection by attack, with uncertainty and length controls |
| Quality and fidelity analysis | Human review plus declared automatic quality metrics |
| Comparative study | Comparable false-positive operating points across methods |
| Final method recommendation | Useful operating range, costs and remaining failure cases |

# Current conclusion

## Working insertion and detection.
## Observed failure under rewriting.

The next contribution is a measured answer to when the signal survives, at what quality cost, and under which attacks.

A research prototype with a clear next experiment

# References: foundations


**[R1] Kirchenbauer et al. (ICML 2023)**

A Watermark for Large Language Models

proceedings.mlr.press/v202/kirchenbauer23a.html

**[R2] Kirchenbauer et al. (ICLR 2024)**

On the Reliability of Watermarks for Large Language Models

ICLR 2024 proceedings. Full source URL in speaker notes.

**[R3] Sadasivan et al. (TMLR, revised 2025)**

Can AI-Generated Text be Reliably Detected?

arxiv.org/abs/2303.11156


Literature supporting the baseline and planned comparisons

# References: further research


**[R4] Kuditipudi et al. (TMLR 2024)**

Robust Distortion-free Watermarks for Language Models

arxiv.org/abs/2307.15593

**[R5] Liu et al. (ICLR 2024)**

A Semantic Invariant Robust Watermark for Large Language Models

arxiv.org/abs/2310.06356

**[R6] Dathathri et al. (Nature 2024)**

Scalable watermarking for identifying large language model outputs

doi.org/10.1038/s41586-024-08025-4

**[R7] Tu et al. (ACL 2024)**

WaterBench: Towards Holistic Evaluation of Watermarks for Large Language Models

aclanthology.org/2024.acl-long.83/


Literature supporting the baseline and planned comparisons